



Beyond CPU and Memory: Event-driven Autoscaling with KEDA

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(he/him)

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About me

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Agenda

Contexts

Why CPU and Memory are late indicator?

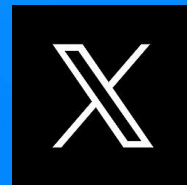
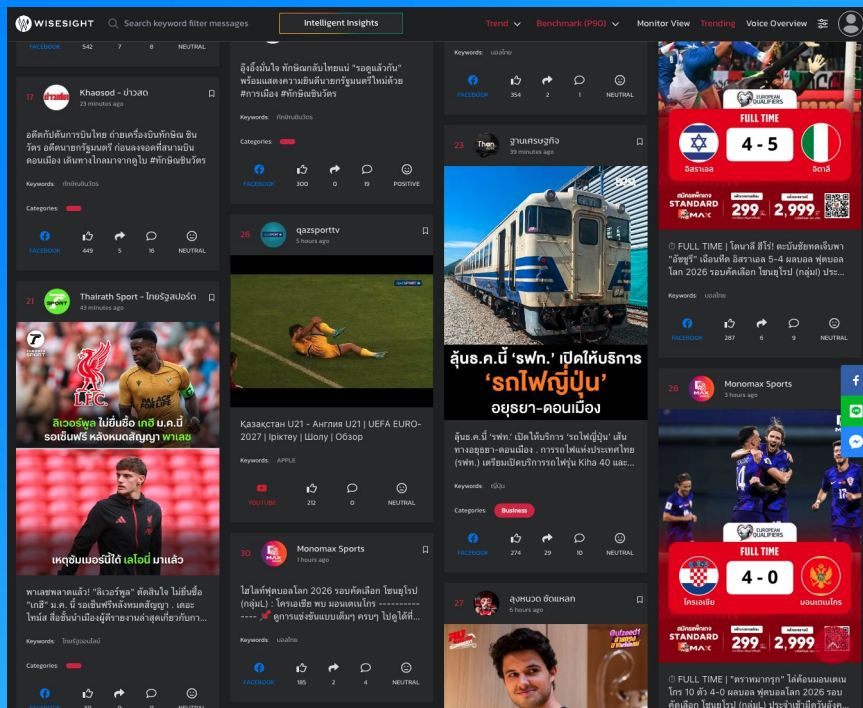
KEDA overview

Use cases

Demo



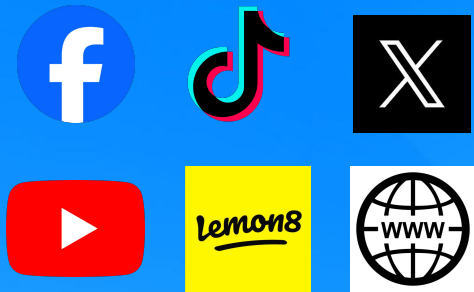
Contexts



Contexts

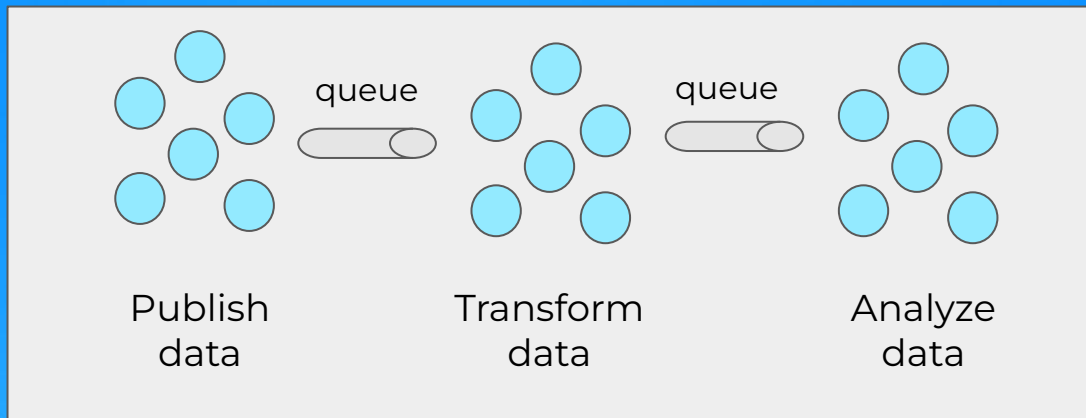


Sources



Handle 3 millions of
messages daily

Kubernetes



Need scalable, resilient and cost effective system



Why CPU & Memory are late indicator?

Staff as
workload
(CPU &
Memory)



<https://fastapi.tiangolo.com/async/#concurrent-burgers>

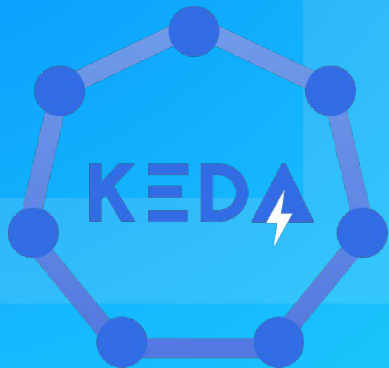


Why CPU & Memory are late indicator?



<https://fastapi.tiangolo.com/async/#concurrent-burgers>





KEDA overview

Kubernetes Event-driven Autoscaling

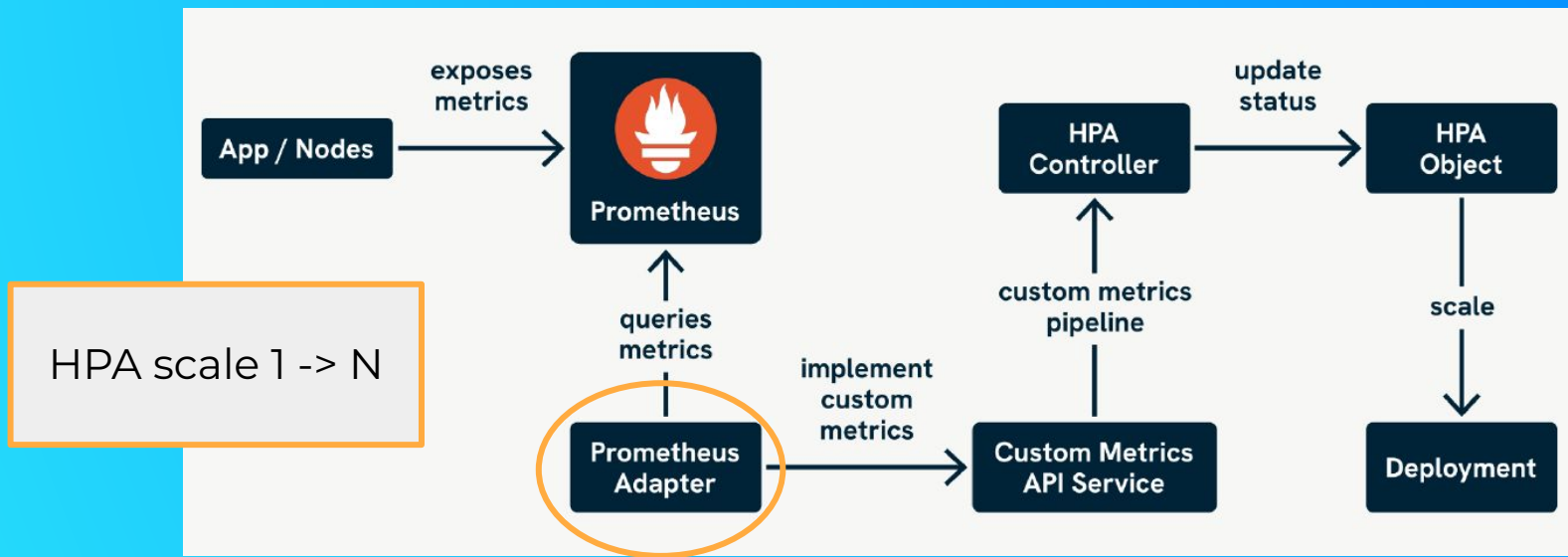
Application autoscaling made simple

- Opensource
- CNCF project (Graduated 2023)
- Simple autoscaling with external metrics
- [Support various scalars](#) (70+ build-in third parties)
- Scale-to-zero

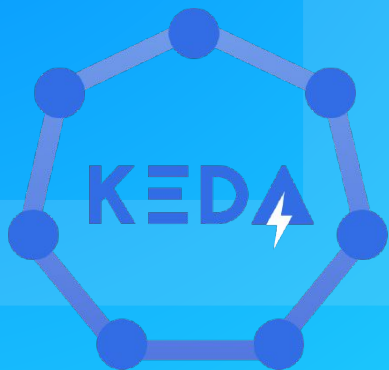


Comparison with HPA

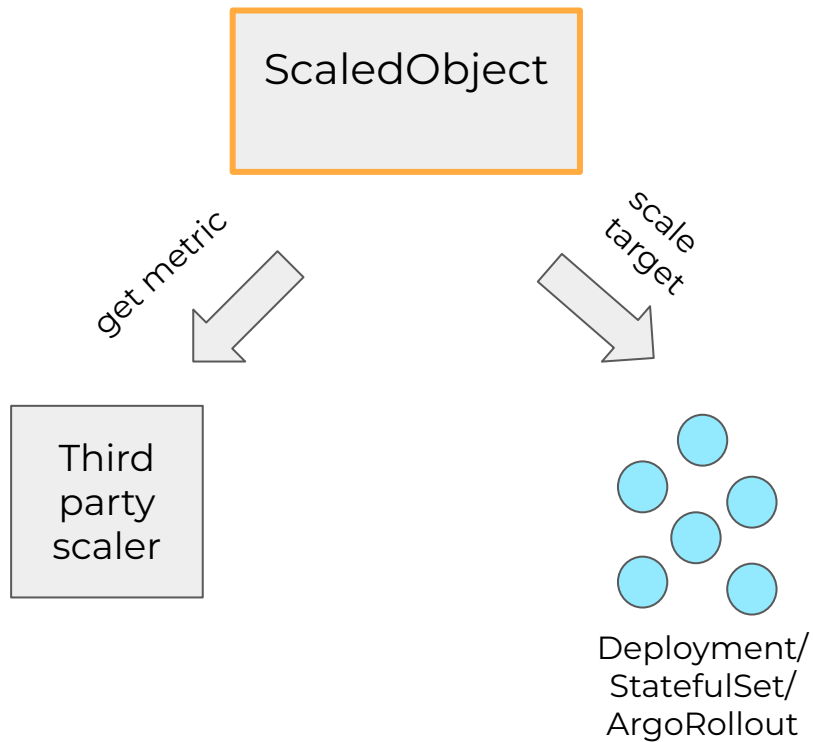
HPA can support Custom metrics

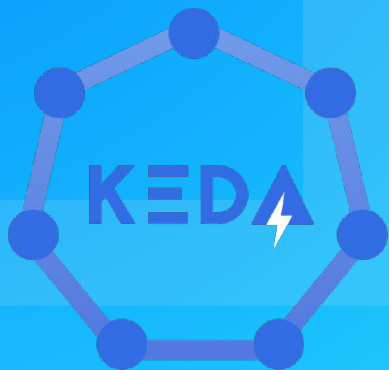


<https://livwyer.io/blog/set-up-kubernetes-scaling-via-prometheus-custom-metrics/>

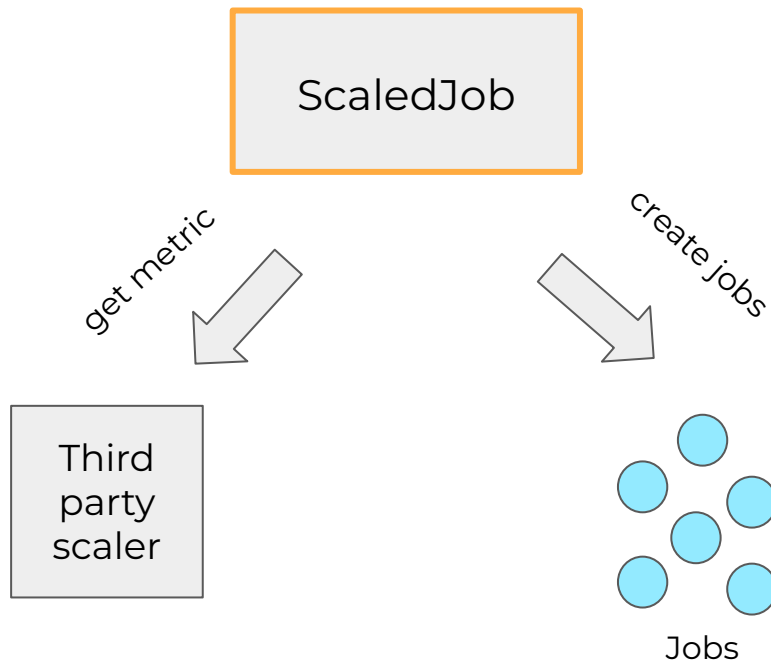


KEDA overview





KEDA overview



Use Cases

Queue-based Scaling

- Number of messages in queue - RabbitMQ, AWS SQS, Kafka

HTTP-based Scaling

- Request rate (RPS) - Prometheus, ServiceMesh
- P90 Latency

Batch job Scaling

- ScaledJob trigger from event query - MongoDB, MySQL, PostgreSQL

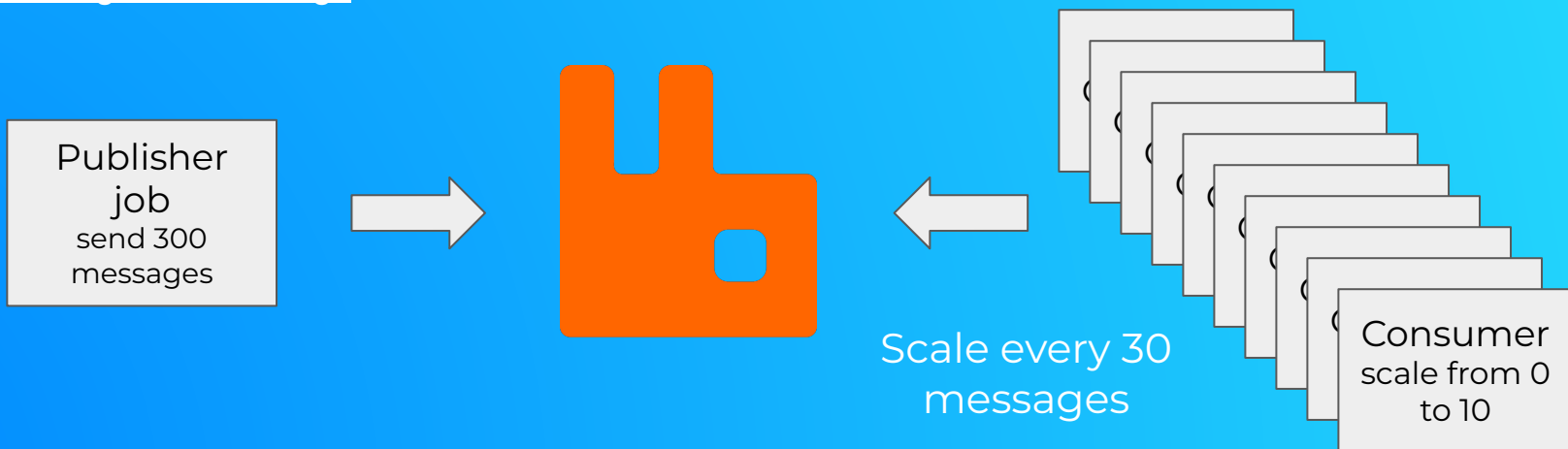




CLOUD NATIVE
COMPUTING FOUNDATION

[https://github.com/songponlekpetch/
beyond-cpu-and-memory-event-dri
ven-autoscaling-with-KEDA.git](https://github.com/songponlekpetch/beyond-cpu-and-memory-event-driven-autoscaling-with-KEDA.git)

Demo



Key takeaway

- CPU/Memory are late indicator
- KEDA extend HPA abilities for autoscaling with event-driven metrics
- Scale-to-zero
- Best practice with Trigger Authentication, HPA Behavior





Q&A